

Linear Regression Analysis

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Exerce-1. Applications of Linear Regression.

List 3 possible applications of Linear Regression (not seen in class). Justify your answer.

Exerce-2. Predict final score using linear regression.

This following table shows the midterm and final scores obtained for students in a course.

X(midterm-score)	Y(final-score)
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
88	74
81	90

Answer the following questions.

a) Plot the data. Do X and Y seem to have linear relationship?

b) Use the method of least squares to find an equation for the prediction of a student's final exam score based on the student's midterm score in the course. Show the linear regression equation.

c) Predict the final exam score of a student who received an 86 on the midterm exam.

Ex-3. Evaluate regression model.

This table shows the predictions made using regression model

Sequence number	x1	x2	x3	Actual y	Predicted y (or ŷ)
1	1	1	1	9	11
2	2	1	1	12	13
3	1	1	3	17	15

Show steps of calculating error using MAE (Mean Absolute Error) and MAPE (Mean Absolute Percentage Error).